



ChainProof

Metric Trust and Reconciliation for Supply-Chain AI

Snowflake-native

CoCo CLI-assisted

One KPI name. Three valid calculations. One governed enterprise answer.

Planning

Material Availability

95%

Procurement

Accepted Supplier Fill

85%

Logistics

On-Time Arrival

90%



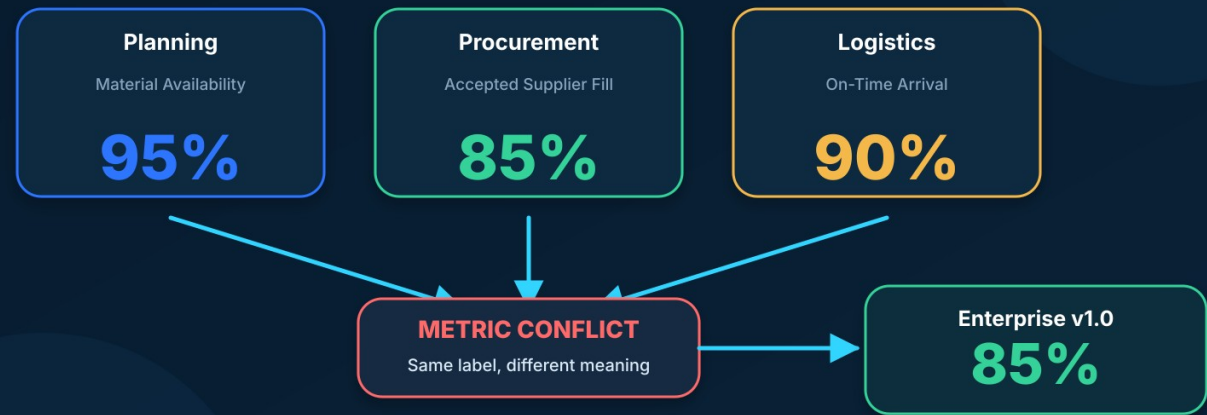
**Enterprise Supplier Fill Rate v1.0 =
85%**

The business problem

Correct data and valid SQL can still produce the wrong business answer.

One KPI name. Three valid calculations.

ChainProof governs the disagreement before AI answers.



Detect → Compare → Assess impact → Evidence → Approve → Publish → Verify

Executive inconsistency

Different teams publish incompatible KPIs under one label.

Supplier disputes

Date and quality rules change accountability.

AI risk

A copilot can confidently select the wrong definition.

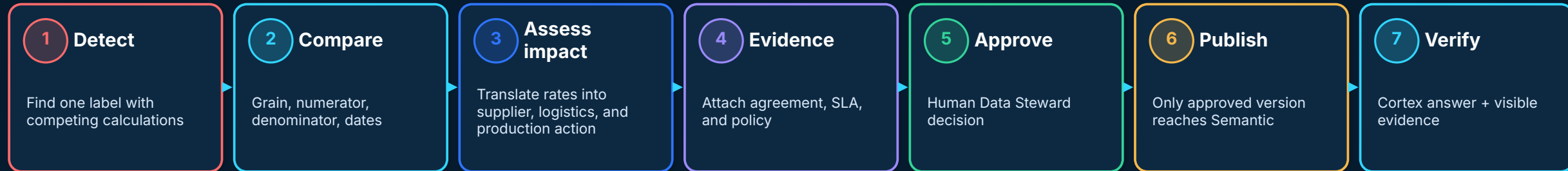
Manual reconciliation

Analysts repeatedly rebuild the same explanation.

Correct SQL + wrong metric meaning = confidently wrong AI

The ChainProof trust lifecycle

AI analytics begins only after the metric is governed.



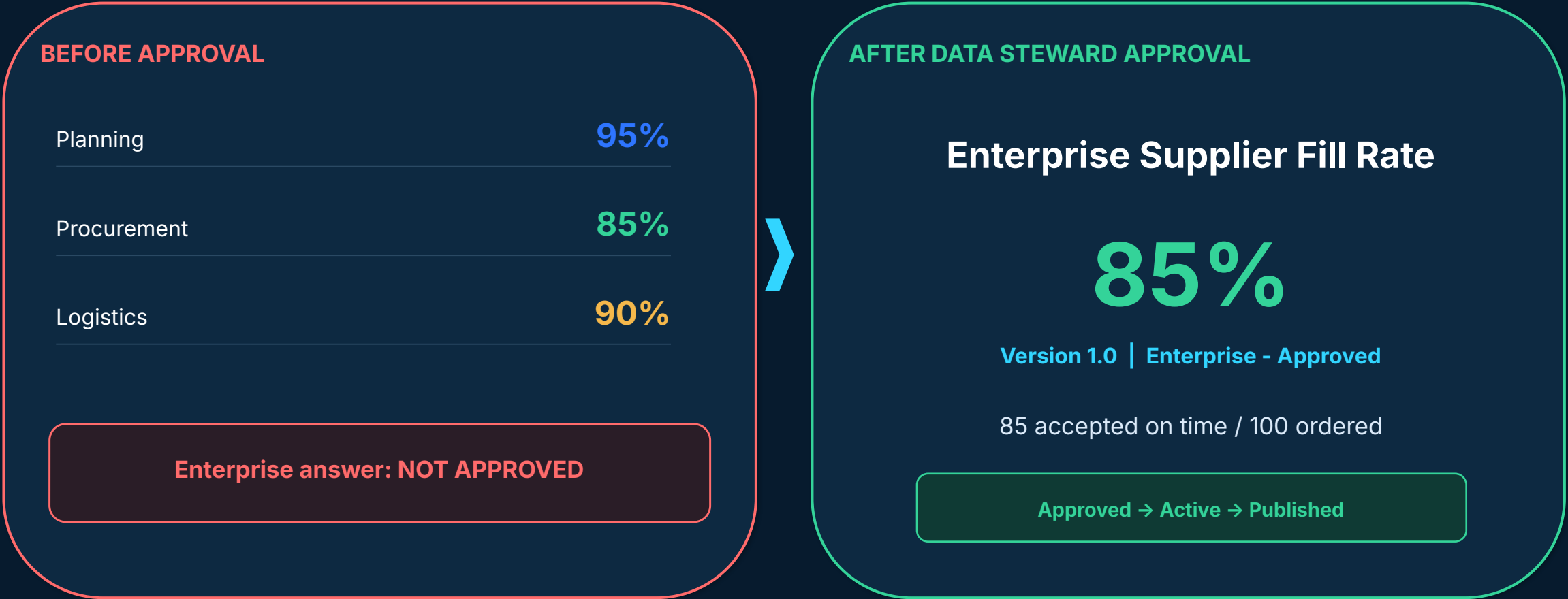
Metric trust firewall

A normal copilot answers the question. ChainProof first proves which definition it is allowed to use.

USP: Semantic layers govern agreed metrics. ChainProof governs the disagreement that exists before agreement.

Human governance changes the answer

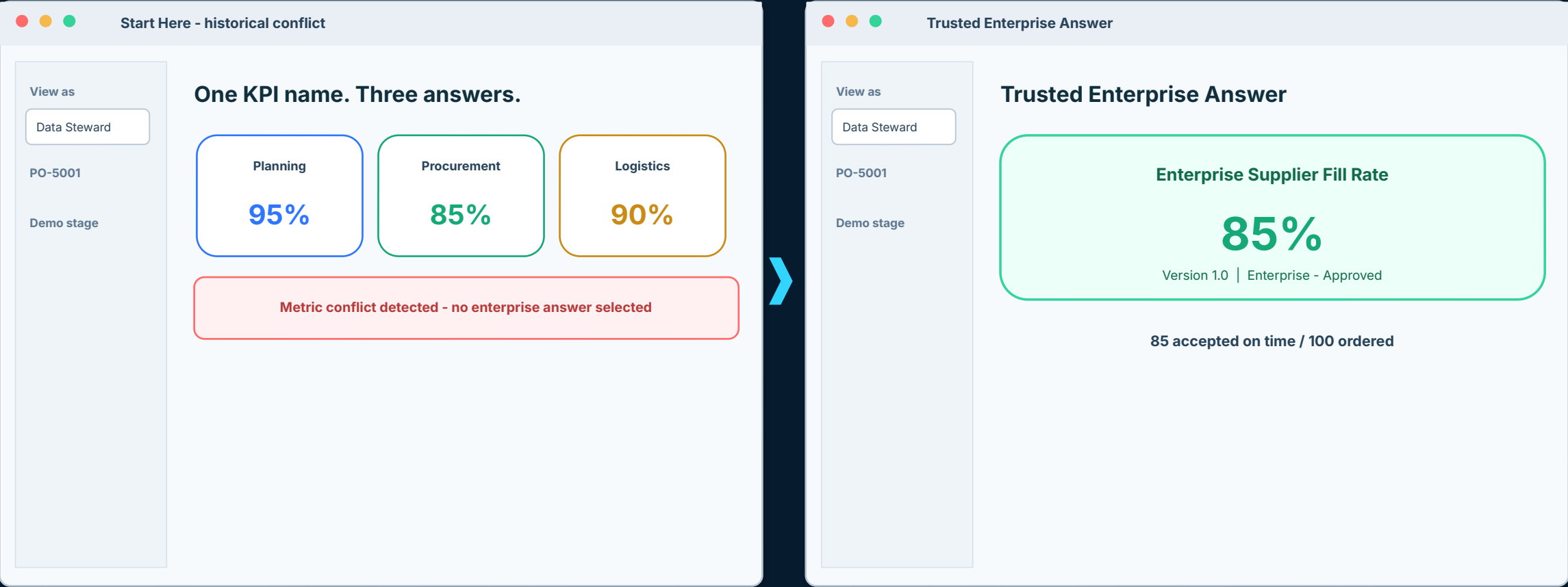
The AI consumes the Data Steward decision; it does not make it.



“View as” demonstrates persona presentation under one learner role. It never changes permissions or formulas.

Judge-first product journey

The deployed app tells the business story before showing architecture.



Start with the bad state

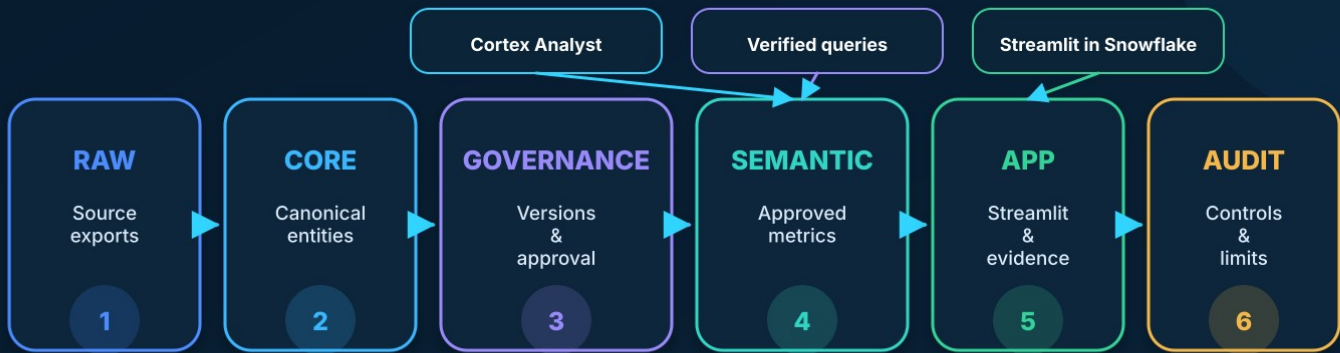
End with the governed answer

Snowflake-native architecture

Each layer has one responsibility and a deterministic validation gate.

Snowflake-native trust architecture

Operational facts, metric governance, AI analytics, and audit remain separated.



Core rule
Only an approved metric version may become a trusted AI answer.

- Semantic View** Only approved metrics are public
- Cortex Analyst** Natural-language governed SQL
- Streamlit in Snowflake** No external app credential required
- Evidence workflow** Citations + trusted boundary
- AUDIT** Release controls + limitations

RAW → CORE → GOVERNANCE → SEMANTIC → APP → AUDIT

Trust controls around every AI answer

ChainProof does not trust generated SQL merely because it compiles.

1

Question scope

PO-specific vs enterprise aggregate

2

Read-only SQL

One SELECT/WITH statement only

3

Semantic boundary

No RAW, CORE, or GOVERNANCE access

4

Version passport

Name, scope, owner, version, formula

5

Evidence trust

Citations; injection fixture excluded

6

Human authority

Advisor cannot approve or write

Fail visibly. Preserve the approved formula. Never fabricate a capability or answer.

Evidence and operational impact

The governed number is explainable and connected to a concrete business action.

View as

Data Steward

PO-5001

Demo stage

Evidence-backed review

Supplier agreement [cited]

Carrier SLA [cited]

Quality policy [cited]

Governance policy [cited]

Advisor cannot approve or write governance data

PO-5001 business impact

Enterprise Supplier Fill Rate

85%

against 90% threshold

15 units

supplier shortfall

Procurement

15 acceptable units short by original PO date

Logistics

10 units arrived after carrier commitment

Planning

5 usable batteries short; up to 5 laptops at risk

Action: supplier escalation + carrier follow-up + production mitigation

One governed metric, three clearly separated operational consequences.

Technical execution and validation

A twelve-part build with deterministic gates, real account evidence, and truthful limitations.



RAW files, rows, edge cases

CORE types, lineage, quality

GOVERNANCE versions, approval, activation

SEMANTIC metrics, relationships, verified SQL

APP scope, persona, evidence, SQL safety

AUDIT release controls and limitations

Known limitation: official Analyst batch evaluation needs account-level task privileges. Live Cortex Analyst and deterministic tests passed.

Why ChainProof is different

It governs metric disagreement - the gap before a normal semantic layer can help.

Category	Typical capability	ChainProof addition
BI dashboard	Displays configured KPIs	Detects conflicting KPI meaning
Data catalog	Documents names and owners	Executes formulas and compares impact
Semantic layer	Publishes agreed definitions	Governs disagreement before agreement
Generic AI copilot	Answers with available metadata	Refuses ambiguity until governed
Metric store	Centralizes formulas	Preserves conflict, approval, activation, rollback

Business value: fewer conflicting reports, faster reconciliation, stronger supplier accountability, and auditable AI answers.



ChainProof

Prevents AI from confidently answering the wrong KPI.

One KPI name. Three valid calculations. One governed enterprise answer.

GitHub

<https://github.com/pankajadam-tech/chainproof-snowflake>

Live app

https://app.snowflake.com/SFEDU05/nxb07453/#/streamlit-apps/CHAINPROOF.APP.CHAINPROOF_APP

Walkthrough

https://drive.google.com/file/d/1sbCqeRjcNWpvU3oHDjV-CBssD_QQyQaK/view?usp=sharing

Snowflake-native • CoCo CLI-assisted • Human governed • Evidence backed